

Java Basics – Learning Guide

1. Introduction to Java

Java is a high-level, object-oriented programming language. Java programs are compiled into bytecode that runs on the Java Virtual Machine (JVM). This makes Java applications portable across systems that have a compatible JVM.

2. Variables and Data Types

A variable stores a value. Common primitive data types are int, long, float, double, char, boolean, byte, and short. String is a class, not a primitive data type.

Example: `int age = 23; String name = "Karthik";`

3. Operators

Java provides arithmetic operators such as +, -, *, /, and %. Relational operators include ==, !=, >, <, >=, and <=. Logical operators include &&, ||, and !.

4. Conditional Statements

The if statement executes code when a condition is true. else provides an alternative. switch can select one case from multiple alternatives.

Example: `if (marks >= 40) { System.out.println("Pass"); } else { System.out.println("Fail"); }`

5. Loops

The for loop is useful when the number of iterations is known. The while loop checks its condition before each iteration. The do-while loop executes its body at least once before checking the condition.

6. Methods

A method is a block of code that performs a task. A Java program commonly starts execution from public static void main(String[] args). Methods can accept parameters and return values.

7. Objects and the new Keyword

A class defines the structure and behavior of objects. The new keyword creates an object.

Example: `Student s = new Student();`

8. Quick Review

Remember: main() is the usual entry point, String is not primitive, new creates objects, and do-while executes at least once.